Module code: BIO311
Module title: Modeling for Computational Biology
Semester 2, 2023/24, Year (4)

Final Examination Paper

Duration: 2 hours

Module Examiner: XXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXXXX

INSTRUCTIONS to CANDIDATES

1. All examination materials must be returned to the invigilator upon completion of the exam. Failure to do so will be deemed as academic misconduct and will be dealt with according to University's policy.

2. The total number of marks is 100. The numbers in brackets indicate the marks assigned to a section, question or sub-question.

3. You need to answer **ALL** questions and all parts.

4. You need to motivate your answer when explicitly required.

5. Answers should be written in the *Answer Booklet* provided.

6. Answers should be written using pen, black or blue, **NO PENCIL**.

7. Only solutions written in English are accepted.

8. You are allowed to use a calculator (but not one on a phone).

Question 1 (30 marks)

a) The binding free energy of a ligand (small molecule A) to a protein can be described by the formula $\Delta G = \Delta H - T\Delta S$. Explain what ΔG, ΔH and ΔS are in this context and how they can be interpreted at the microscopic level.

[20 marks]

b) Assume there is a second ligand (small molecule B) with binding affinity $\Delta G_B < \Delta G_A$. Explain how the two molecules will compete for binding to the protein if they are added to the solution at the same concentration.

[5 marks]

c) Which of the two molecules in question (b) is a better drug candidate if we consider the binding affinity alone? Motivate your answer.

[5 marks]

Question 2 (30 marks)

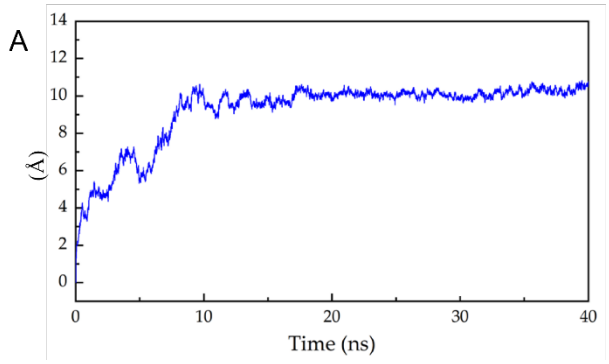
a) Explain what RMSD and RMSF are in the context of molecular dynamics simulations.

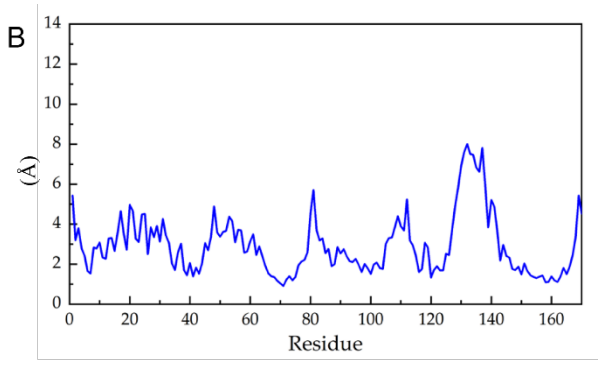
$$RMSD(t) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{r}_i(t) - \boldsymbol{r}_i(0))^2}$$

$$RMSF(i) = \sqrt{\frac{1}{T}\sum_{t=0}^{T}\left(\boldsymbol{r}_i(t) - \boldsymbol{r}_{i,AVE}\right)^2}$$

[12 marks]

b) The following two graphs (A and B) represent the RMSD and RMSF of a 40 ns simulation of a protein system in water. They are both computed by considering the protein's alpha carbons only. Identify which graph corresponds to the RMSD and which to the RMSF, and explain what the x and y axes represent.





[5 marks]

c) From the RMSD plot, identify the region after which the simulation can be considered equilibrated and motivate your answer.

[4 marks]

d) From the RMSF plot, identify which part of the protein is more likely to be a loop and motivate your answer.

[4 marks]

e) Based on the results presented, do you think it is necessary to extend the duration of the simulation to 100 ns before performing further data analysis? Motivate your answer.

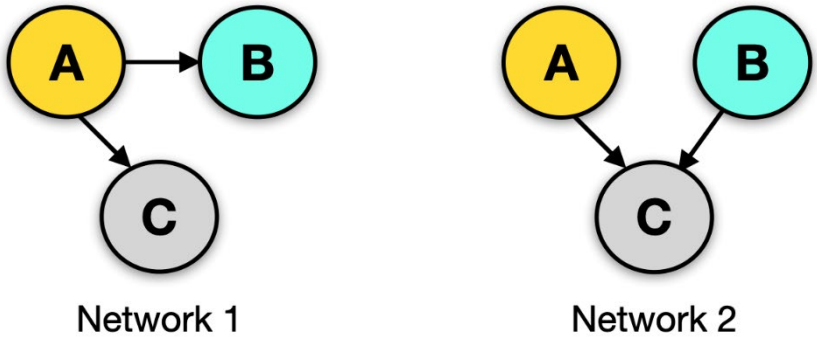
[5 marks]

Question 3 (40 marks)

a) Explain Bayesian networks in the context of molecular pathways and how such networks are typically built.

[14 marks]

b) To understand the causal relationships between three proteins in a molecular pathway, two different Bayesian networks were constructed, where each regulator is modeled by a binary (Bernoulli) random variable.



Network 1 Network 2

Write down the minimum number of parameters needed to specify the joint distribution P(A,B,C) under the following independence assumptions:

(i) A, B, C are mutually independent
(ii) A, B, C follow the independence assumptions of **Network 2** above
(iii) No independencies

[6 marks]

c) Given the data summarized in the three contingency tables, compute the MLE for the model parameters $\theta_1, \theta_2, \theta_3, \theta_4, \theta_5$ in Network 1 of point (b).

```
> table(A)
A
Off  On
 30  70
```

```
> table(A,B)
     B
A     Off On
  Off   2 28
  On   58 12
```

```
> table(A,C)
     C
A     Off On
  Off  28  2
  On   34 36
```

$$P(A = On) = \theta_1;\ P(B = On|A) = \begin{cases} \theta_2, A = On \\ \theta_3, A = Off \end{cases};\quad P(C = On|A) = \begin{cases} \theta_4, A = On \\ \theta_5, A = Off \end{cases}$$

[8 marks]

d) With the computed MLEs in question 4b, calculate the following probabilities:

(i) P(A = On, B = On, C = On)
(ii) P(A = On| B = On)

[6 marks]

e) Assume the log-likelihood evaluated at the MLEs for network 1 and network 2 of point (b) are −144.83 and −143.10 respectively. Compute their BIC scores with n = 100. Judging from their BIC, which network structure is more preferred?

[6 marks]